

MODULE END MY SQL

E-Commerce Customer Churn Analysis

Introduction

Customer churn is one of the major challenges faced by e-commerce businesses. Understanding why customers stop purchasing helps organizations improve customer satisfaction and retention. This project analyses customer data using MySQL to identify churn patterns and customer behaviour.

The dataset contains information such as customer demographics, shopping preferences, payment methods, satisfaction scores, and order history. Through data cleaning, transformation, and analysis, the project aims to uncover valuable insights related to customer churn. The findings can help e-commerce companies make data-driven decisions and develop effective strategies to improve customer retention and business growth.

Dataset Description

The dataset contains customer information, including:

- Customer ID
- Tenure
- Preferred Login Device
- Preferred Payment Mode
- Preferred Order Category
- City Tier
- Gender
- Satisfaction Score
- Warehouse to Home Distance
- Cashback Amount
- Order Count
- Complaint Status
- Churn Status

Database ecomm is created in MySQL

```
1
2 • CREATE DATABASE ecomm;
3 • USE ecomm;
4
5 • CREATE TABLE customer_churn(
6     CustomerID          INT    PRIMARY KEY,
7     Churn                BIT,
8     Tenure               INT,
9     PreferredLoginDevice VARCHAR(20),
10    CityTier              INT,
11    WarehouseToHome      INT,
12    PreferredPaymentMode  VARCHAR(20),
13    Gender                ENUM('Male','Female'),
14    HourSpendOnApp        INT,
15    NumberOfDeviceRegistered INT,
16    PreferredOrderCat     VARCHAR(20),
17    SatisfactionScore     INT,
18    MaritalStatus         VARCHAR(10),
19    NumberOfAddress       INT,
20    Complain              BIT,
21    OrderAmountHikeFromLastYear INT,
22    CouponUsed            INT,
23    OrderCount            INT,
24    DaySinceLastOrder     INT,
25    CashbackAmount        INT
26 );
```

```
• INSERT INTO customer_churn(CustomerID,Churn,Tenure,PreferredLoginDevice,CityTier,WarehouseToHome,PreferredPaymentMode,Gender,HourSpendOnApp,NumberOfDeviceRegistered,PreferredOrderCat,SatisfactionScore,MaritalStatus,NumberOfAddress)
(50001,1,4,'Mobile Phone',3,6,'Debit Card','Female',3,3,'Laptop & Accessory',2,'Single',9,1,11,1,1,5,160),
(50002,1,NULL,'Phone',1,8,'UPI','Male',3,4,'Mobile',3,'Single',7,1,15,0,1,0,121),
(50003,1,NULL,'Phone',1,30,'Debit Card','Male',2,4,'Mobile',3,'Single',6,1,14,0,1,3,120),
(50004,1,0,'Phone',3,15,'Debit Card','Male',2,4,'Laptop & Accessory',5,'Single',8,0,23,0,1,3,134),
(50005,1,0,'Phone',1,12,'CC','Male',NULL,3,'Mobile',5,'Single',3,0,11,1,1,3,130),
(50006,1,0,'Computer',1,22,'Debit Card','Female',3,5,'Mobile Phone',5,'Single',2,1,22,4,6,7,139),
(50007,1,NULL,'Phone',3,11,'Cash on Delivery','Male',2,3,'Laptop & Accessory',2,'Divorced',4,0,14,0,1,0,121),
(50008,1,NULL,'Phone',1,6,'CC','Male',3,3,'Mobile',2,'Divorced',3,1,16,2,2,0,123),
(50009,1,13,'Phone',3,9,'E wallet','Male',NULL,4,'Mobile',3,'Divorced',2,1,14,0,1,2,127),
(50010,1,NULL,'Phone',1,31,'Debit Card','Male',2,5,'Mobile',3,'Single',2,0,12,1,1,1,123),
(50011,1,4,'Mobile Phone',1,18,'Cash on Delivery','Female',2,3,'Others',3,'Divorced',2,0,NULL,9,15,0,295),
(50012,1,11,'Mobile Phone',1,6,'Debit Card','Male',3,4,'Fashion',3,'Single',10,1,13,0,1,0,154),
(50013,1,0,'Phone',1,11,'COD','Male',2,3,'Mobile',3,'Single',2,1,13,2,2,2,134),
(50014,1,0,'Phone',1,15,'CC','Male',3,4,'Mobile',3,'Divorced',1,1,17,0,1,0,134),
(50015,1,9,'Mobile Phone',3,15,'Credit Card','Male',3,4,'Fashion',2,'Single',2,0,16,0,4,7,196),
(50016,1,NULL,'Phone',2,12,'UPI','Male',3,3,'Mobile',5,'Married',5,1,22,1,1,2,121),
(50017,1,0,'Computer',1,12,'Debit Card','Female',NULL,4,'Mobile',2,'Single',2,1,18,1,1,0,129),
(50018,1,0,'Mobile Phone',3,11,'E wallet','Male',2,4,'Laptop & Accessory',3,'Single',2,1,11,1,1,3,157),
(50019,1,0,'Computer',1,13,'Debit Card','Male',3,5,'Laptop & Accessory',3,'Single',2,1,24,1,1,6,161),
(50020,1,19,'Mobile Phone',1,20,'Debit Card','Female',3,3,'Mobile Phone',4,'Divorced',10,1,18,1,4,3,150),
(50021,1,0,'Mobile Phone',3,12,'Debit Card','Male',3,5,'Laptop & Accessory',3,'Divorced',5,1,18,6,7,7,162),
(50022,1,20,'Mobile Phone',1,29,'Credit Card','Female',3,3,'Fashion',2,'Divorced',2,0,12,11,15,6,203),
(50023,1,NULL,'Mobile Phone',3,28,'E wallet','Male',2,3,'Mobile Phone',3,'Single',2,1,19,0,1,0,117),
(50024,1,0,'Phone',3,26,'Debit Card','Female',3,5,'Laptop & Accessory',3,'Divorced',4,1,11,1,2,2,146),
(50025,1,14,'Computer',1,14,'Debit Card','Male',2,5,'Fashion',2,'Single',3,0,15,0,1,3,207),
```

Result Grid															Filter Rows:										Edit:										Export/Import:										Wrap Cell Content:										Fetch rows:									
	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress																																																		
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single	9																																																		
	50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3	Single	7																																																		
	50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3	Single	6																																																		
	50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single	8																																																		
	50005	1	0	Phone	1	12	CC	Male	NULL	3	Mobile	5	Single	3																																																		
	50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5	Single	2																																																		
	50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2	Divorced	4																																																		
	50008	1	NULL	Phone	1	6	CC	Male	3	3	Mobile	2	Divorced	3																																																		
	50009	1	13	Phone	3	9	E wallet	Male	NULL	4	Mobile	3	Divorced	2																																																		
	50010	1	NULL	Phone	1	31	Debit Card	Male	2	5	Mobile	3	Single	2																																																		
	50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3	Divorced	2																																																		
	50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion	3	Single	10																																																		
	50013	1	0	Phone	1	11	COD	Male	2	3	Mobile	3	Single	2																																																		
	50014	1	0	Phone	1	15	CC	Male	3	4	Mobile	3	Divorced	1																																																		
	50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion	2	Single	2																																																		
	50016	1	NULL	Phone	2	12	UPI	Male	3	3	Mobile	5	Married	5																																																		
	50017	1	0	Computer	1	12	Debit Card	Female	NULL	4	Mobile	2	Single	2																																																		
	50018	1	0	Mobile Phone	3	11	E wallet	Male	2	4	Laptop & Accessory	3	Single	2																																																		
	50019	1	0	Computer	1	13	Debit Card	Male	3	5	Laptop & Accessory	3	Single	2																																																		
	50020	1	19	Mobile Phone	1	20	Debit Card	Female	3	3	Mobile Phone	4	Divorced	10																																																		

Data Cleaning:

❖ Handling Missing Values and Outliers:

➤ Impute mean for the following columns and round off to the nearest integer

1) WarehouseToHome

```
UPDATE customer_churn
SET WarehouseToHome = (
  SELECT ROUND(AVG(WarehouseToHome))
  FROM (
    SELECT WarehouseToHome
    FROM customer_churn
    WHERE WarehouseToHome IS NOT NULL
  ) t
)
WHERE WarehouseToHome IS NULL;
```

Filter Rows:

CityTier	WarehouseToHome
	6
	8
	30
	15
	12
	22
	11
	6
	9
	31
	18
	6
	11
	15
	15

2) HourSpendOnApp

```
UPDATE customer_churn
SET HourSpendOnApp = (
  SELECT ROUND(AVG(HourSpendOnApp))
  FROM (
    SELECT HourSpendOnApp
    FROM customer_churn
    WHERE HourSpendOnApp IS NOT NULL
  ) t
)
WHERE HourSpendOnApp IS NULL;
```

Rows:	
ler	HourSpendOnApp
e	3
	3
	2
	2
	3
e	3
	2
	3
	3
	2
e	2
	3
	2
	3
	3
	3

3) OrderAmountHikeFromlastYear

		OrderAmountHikeFromlastYear
<pre> UPDATE customer_churn SET OrderAmountHikeFromlastYear = (SELECT ROUND(AVG(OrderAmountHikeFromlastYear)) FROM (SELECT OrderAmountHikeFromlastYear FROM customer_churn WHERE OrderAmountHikeFromlastYear IS NOT NULL) t) WHERE OrderAmountHikeFromlastYear IS NULL; </pre>		11
		15
		14
		23
		11
		22
		14
		16
		14
		12
		16
		13
		13

4) DaySinceLastOrder

		DaySinceLastOrder
<pre> UPDATE customer_churn SET DaySinceLastOrder = (SELECT ROUND(AVG(DaySinceLastOrder)) FROM (SELECT DaySinceLastOrder FROM customer_churn WHERE DaySinceLastOrder IS NOT NULL) t) WHERE DaySinceLastOrder IS NULL; </pre>		5
		0
		3
		3
		3
		7
		0
		0
		2
		1
		8
		0

➤ Impute mode for the following columns:

1) Tenure

```
UPDATE customer_churn
SET Tenure = (
    SELECT Tenure
    FROM (
        SELECT Tenure
        FROM customer_churn
        GROUP BY Tenure
        ORDER BY COUNT(*) DESC
        LIMIT 1
    ) t
)
WHERE Tenure IS NULL;
```

Tenure
4
1
1
0
0
0
1
1
13
1

2) CouponUsed

```
UPDATE customer_churn
SET CouponUsed =
(
    SELECT CouponUsed
    FROM (
        SELECT CouponUsed
        FROM customer_churn
        GROUP BY CouponUsed
        ORDER BY COUNT(*) DESC
        LIMIT 1
    ) t
)
WHERE CouponUsed IS NULL;
```

3) OrderCount

<pre>UPDATE customer_churn SET OrderCount = (SELECT OrderCount FROM (SELECT OrderCount FROM customer_churn GROUP BY OrderCount ORDER BY COUNT(*) DESC LIMIT 1) t) WHERE OrderCount IS NULL;</pre>	<table><tr><th>OrderCount</th></tr><tr><td>1</td></tr><tr><td>1</td></tr><tr><td>1</td></tr><tr><td>1</td></tr><tr><td>1</td></tr><tr><td>6</td></tr><tr><td>1</td></tr><tr><td>2</td></tr><tr><td>1</td></tr><tr><td>1</td></tr><tr><td>1</td></tr><tr><td>15</td></tr><tr><td>-</td></tr></table>	OrderCount	1	1	1	1	1	6	1	2	1	1	1	15	-
OrderCount															
1															
1															
1															
1															
1															
6															
1															
2															
1															
1															
1															
15															
-															

➤ Outlier Treatment

Rows having WarehouseToHome values greater than 100 were identified as outliers and removed from the dataset

----- removing outliers -----

```
DELETE FROM customer_churn
WHERE WarehouseToHome > 100;
```

❖ Dealing with Inconsistencies:

➤ Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferredOrderCat' column with “Mobile Phone” to ensure uniformity.

```
5762 • UPDATE customer_churn
5763 SET PreferredLoginDevice = 'Mobile Phone'
5764 WHERE PreferredLoginDevice = 'Phone';
5765
5766 • select * from customer_churn;
```

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome
50001	1	4	Mobile Phone	3	6
50002	1	1	Mobile Phone	1	8
50003	1	1	Mobile Phone	1	30
50004	1	0	Mobile Phone	3	15
50005	1	0	Mobile Phone	1	12
50006	1	0	Computer	1	22
50007	1	1	Mobile Phone	3	11

```
5768 • UPDATE customer_churn
5769 SET PreferredOrderCat = 'Mobile Phone'
5770 WHERE PreferredOrderCat = 'Mobile';
```

Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus
Female	3	3	Laptop & Accessory	2	Single
Male	3	4	Mobile Phone	3	Single
Male	2	4	Mobile Phone	3	Single
Male	2	4	Laptop & Accessory	5	Single
Male	3	3	Mobile Phone	5	Single
Female	3	5	Mobile Phone	5	Single
Male	2	3	Laptop & Accessory	2	Divorced
Male	3	3	Mobile Phone	2	Divorced
Male	3	4	Mobile Phone	3	Divorced

customer_churn 2

➤ Standardize payment mode values:

1) Replace "COD" with "Cash on Delivery"

```
5777 • UPDATE customer_churn
5778 SET PreferredPaymentMode = 'Credit Card'
5779 WHERE PreferredPaymentMode = 'CC';
```

Tier	WarehouseToHome	PreferredPaymentMode	Gender
	11	Cash on Delivery	Male
	6	Credit Card	Male
	9	E wallet	Male
	31	Debit Card	Male
	18	Cash on Delivery	Female
	6	Debit Card	Male
	11	Cash on Delivery	Male
	15	Credit Card	Male
	15	Credit Card	Male

2) "CC" with "Credit Card" in the PreferredPaymentMode column

```
5773 • UPDATE customer_churn
5774     SET PreferredPaymentMode = 'Cash on Delivery'
5775     WHERE PreferredPaymentMode = 'COD';
5776
```

ier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendO
	15	Debit Card	Male	2
	12	Credit Card	Male	3
	22	Debit Card	Female	3
	11	Cash on Delivery	Male	2
	6	Credit Card	Male	3
	9	E wallet	Male	3
	31	Debit Card	Male	2
	18	Cash on Delivery	Female	2
	6	Debit Card	Male	3

Data Transformation

❖ Column Renaming:

➤ Rename the column "PreferedOrderCat" to "PreferredOrderCat".

```
5782 • ALTER TABLE customer_churn
5783     CHANGE PreferedOrderCat PreferredOrderCat VARCHAR(100);
5784
5785 • select * from customer_churn;
```

	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfD
▶	6	Debit Card	Female	3	3
	8	UPI	Male	3	4
	30	Debit Card	Male	2	4
	15	Debit Card	Male	2	4
	12	Credit Card	Male	3	3
	22	Debit Card	Female	3	5
	11	Cash on Delivery	Male	2	3
	6	Credit Card	Male	3	3
	9	E wallet	Male	3	4

- Rename the column "HourSpendOnApp" to "HoursSpentOnApp"

```
ALTER TABLE customer_churn
CHANGE HourSpendOnApp HoursSpentOnApp INT;
```

CustomerID	HoursSpentOnApp	NumberOfDeviceRegistered	Product
50001	3	3	Laptop
50002	3	4	Mobile
50003	2	4	Mobile
50004	2	4	Laptop
50005	3	3	Mobile
50006	3	5	Mobile
50007	2	3	Laptop
50008	3	3	Mobile
50009	3	4	Mobile

❖ Creating New Columns:

➤ ComplaintReceived

- Yes → Complain = 1
- No → Complain = 0

```
5792 • ALTER TABLE customer_churn
5793     ADD ComplaintReceived VARCHAR(3);
5794
5795 • UPDATE customer_churn
5796     SET ComplaintReceived =
5797     CASE
5798         WHEN Complain = 1 THEN 'Yes'
5799         ELSE 'No'
5800     END;
5801
5802 • SELECT CustomerID, Complain, ComplaintReceived
5803     FROM customer_churn
5804     LIMIT 10;
```

CustomerID	Complain	ComplaintReceived
50001	1	Yes
50002	1	Yes
50003	1	Yes
50004	0	No
50005	0	No
50006	1	Yes
50007	0	No
50008	1	Yes
50009	1	Yes
50010	0	No

➤ ChurnStatus

- Churned → Churn = 1
- Active → Churn = 0

```
5807 • ALTER TABLE customer_churn
5808     ADD ChurnStatus VARCHAR(10);
5809
5810
5811 • UPDATE customer_churn
5812     SET ChurnStatus =
5813     CASE
5814         WHEN Churn = 1 THEN 'Churned'
5815         ELSE 'Active'
5816     END;
5817
5818 • SELECT CustomerID, Churn, ChurnStatus
5819     FROM customer_churn
5820     LIMIT 10;
```

Result Grid

CustomerID	Churn	ChurnStatus
50001	1	Churned
50002	1	Churned
50003	1	Churned
50004	1	Churned
50005	1	Churned
50006	1	Churned
50007	1	Churned
50008	1	Churned
50009	1	Churned
50010	1	Churned

❖ Dropped Columns

- Drop the columns "Churn" and "Complain" from the table.

```
5823 • ALTER TABLE customer_churn
5824     DROP COLUMN Churn,
5825     DROP COLUMN Complain;
5826
5827 • select * from customer_churn;
```

Result Grid

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseTo
50109	23	Mobile Phone	3	6
50110	0	Mobile Phone	1	16
50111	16	Mobile Phone	3	22
50112	0	Mobile Phone	1	8

Data Exploration and Analysis

1) Count of churned and active customers.

```
5832 • SELECT ChurnStatus, COUNT(*) AS CustomerCount
5833 FROM customer_churn
5834 GROUP BY ChurnStatus;
```

ChurnStatus	CustomerCount
Churned	948
Active	4680

2) Average tenure and total cashback amount of churned customers

```
5838 • SELECT
5839 AVG(Tenure) AS AvgTenure,
5840 SUM(CashbackAmount) AS TotalCashback
5841 FROM customer_churn
5842 WHERE ChurnStatus='Churned';
```

AvgTenure	TotalCashback
3.1762	152030

3) Percentage of churned customers who raised complaints

```
5846 • SELECT
5847 ROUND(
5848 COUNT(CASE WHEN ComplaintReceived='Yes' THEN 1 END)*100.0/
5849 COUNT(*),2
5850 ) AS ComplaintPercentage
5851 FROM customer_churn
5852 WHERE ChurnStatus='Churned';
```

ComplaintPercentage
53.59

4) City tier with the highest number of churned Laptop & Accessory customers

```
5856 • SELECT CityTier, COUNT(*) AS Total
5857 FROM customer_churn
5858 WHERE ChurnStatus='Churned'
5859 AND PreferredOrderCat='Laptop & Accessory'
5860 GROUP BY CityTier
5861 ORDER BY Total DESC
5862 LIMIT 1;
```

CityTier	Total
3	150

5) Most preferred payment mode among active customers

```
5868 • SELECT PreferredPaymentMode, COUNT(*) AS Total
5869 FROM customer_churn
5870 WHERE ChurnStatus='Active'
5871 GROUP BY PreferredPaymentMode
5872 ORDER BY Total DESC
5873 LIMIT 1;
```

PreferredPaymentMode	Total
Debit Card	1956

6) Total order amount hike for single customers using mobile phones.

```
5877 • SELECT SUM(OrderAmountHikeFromLastYear) AS TotalHike
5878 FROM customer_churn
5879 WHERE MaritalStatus='Single'
5880 AND PreferredLoginDevice='Mobile Phone';
```

TotalHike
19593

7) Average registered devices for UPI users

```
5884 • SELECT AVG(NumberOfDeviceRegistered) AS AvgDevices
5885 FROM customer_churn
5886 WHERE PreferredPaymentMode='UPI';
```

Result Grid



 Filter Rows:

Export: 

Wrap Cell Co

	AvgDevices
▶	3.7174

8) City tier with highest customer count

```
5890 • SELECT CityTier, COUNT(*) AS TotalCustomers
5891 FROM customer_churn
5892 GROUP BY CityTier
5893 ORDER BY TotalCustomers DESC
5894 LIMIT 1;
```

CityTier	TotalCustomers
1	3666

9) Gender with highest coupon usage

```
5898 • SELECT Gender,
5899 SUM(CouponUsed) AS TotalCoupons
5900 FROM customer_churn
5901 GROUP BY Gender
5902 ORDER BY TotalCoupons DESC
5903 LIMIT 1;
```

Gender	TotalCoupons
Male	5494

10) Customer count and maximum app hours by order category.

```
5907 • SELECT PreferredOrderCat,
5908 COUNT(*) AS CustomerCount,
5909 MAX(HoursSpentOnApp) AS MaxHours
5910 FROM customer_churn
5911 GROUP BY PreferredOrderCat;
5912
```

PreferredOrderCat	CustomerCount	MaxHours
Laptop & Accessory	2050	5
Mobile Phone	2078	5
Others	264	4
Fashion	826	5
Grocery	410	4

11) Total order count using credit cards with maximum satisfaction score

```
5915 • SELECT SUM(OrderCount) AS TotalOrders
5916 FROM customer_churn
5917 WHERE PreferredPaymentMode='Credit Card'
5918 AND SatisfactionScore =
5919 (
5920 SELECT MAX(SatisfactionScore)
5921 FROM customer_churn
5922 );
```

TotalOrders
1122

12) Average satisfaction score among customers who complained.

```
5926 • SELECT AVG(SatisfactionScore) AS AvgScore
5927 FROM customer_churn
5928 WHERE ComplaintReceived='Yes';
5929
```

AvgScore
2.9988

13) Preferred order categories for customers using more than five coupons

```
5932 • SELECT DISTINCT PreferredOrderCat
5933 FROM customer_churn
5934 WHERE CouponUsed > 5;
5935
```

PreferredOrderCat
Others
Laptop & Accessory
Fashion
Mobile Phone
Grocery

14) Top three categories with highest average cashback

```
5938 • SELECT PreferredOrderCat,  
5939     AVG(CashbackAmount) AS AvgCashback  
5940 FROM customer_churn  
5941 GROUP BY PreferredOrderCat  
5942 ORDER BY AvgCashback DESC  
5943 LIMIT 3;
```

PreferredOrderCat	AvgCashback
Others	304.4545
Grocery	266.2366
Fashion	210.4031

15) Payment modes where average tenure is 10 months and order count exceeds 500

```
5947 • SELECT PreferredPaymentMode  
5948 FROM customer_churn  
5949 WHERE OrderCount > 500  
5950 GROUP BY PreferredPaymentMode  
5951 HAVING AVG(Tenure)=10;
```

PreferredPaymentMode

16) Customer segmentation based on warehouse distance.

```
5958 • SELECT  
5959 CASE  
5960   WHEN WarehouseToHome <= 5 THEN 'Very Close Distance'  
5961   WHEN WarehouseToHome <= 10 THEN 'Close Distance'  
5962   WHEN WarehouseToHome <= 15 THEN 'Moderate Distance'  
5963   ELSE 'Far Distance'  
5964 END AS DistanceCategory,  
5965 ChurnStatus,  
5966 COUNT(*) AS TotalCustomers  
5967 FROM customer_churn  
5968 GROUP BY DistanceCategory, ChurnStatus  
5969 ORDER BY DistanceCategory;
```

DistanceCategory	ChurnStatus	TotalCustomers
Close Distance	Active	1696
Close Distance	Churned	265
Far Distance	Active	1871
Far Distance	Churned	498
Moderate Distance	Active	1106
Moderate Distance	Churned	184
Very Close Distance	Active	7
Very Close Distance	Churned	1

17) Customer details for married customers in City Tier-1 with above-average order counts.

```

5973 • SELECT *
5974 FROM customer_churn
5975 WHERE MaritalStatus='Married'
5976 AND CityTier=1
5977 AND OrderCount >
5978 (
5979 SELECT AVG(OrderCount)
5980 FROM customer_churn
5981 );

```

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceRegistered
50049	3	Computer	1	15	Credit Card	Male	3	3
50119	13	Mobile Phone	1	30	Cash on Delivery	Male	3	3
50132	15	Mobile Phone	1	16	Credit Card	Female	2	4
50133	13	Mobile Phone	1	8	Credit Card	Male	2	3

PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	OrderAmountLikeFromLastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived	ChurnStatus
Laptop & Accessory	5	Married	2	11	0	6	10	131	No	Active
Laptop & Accessory	2	Married	2	17	2	4	7	133	No	Active
Mobile Phone	5	Married	1	11	4	7	8	126	Yes	Active
Others	3	Married	1	16	0	3	9	294	No	Active

Customer Return Analysis

❖ Create a 'customer_returns' table in the 'ecomm' database and insert the following data:

```

• CREATE TABLE customer_returns
(
    ReturnID INT PRIMARY KEY,
    CustomerID INT,
    ReturnDate DATE,
    RefundAmount DECIMAL(10,2)
);

```

```

5997 • INSERT INTO customer_returns
5998 VALUES
5999 (1001,50022,'2023-01-01',2130),
6000 (1002,50316,'2023-01-23',2000),
6001 (1003,51099,'2023-02-14',2290),
6002 (1004,52321,'2023-03-08',2510),
6003 (1005,52928,'2023-03-20',3000),
6004 (1006,53749,'2023-04-17',1740),
6005 (1007,54206,'2023-04-21',3250),
6006 (1008,54838,'2023-04-30',1990);

```

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130.00
1002	50316	2023-01-23	2000.00
1003	51099	2023-02-14	2290.00
1004	52321	2023-03-08	2510.00
1005	52928	2023-03-20	3000.00
1006	53749	2023-04-17	1740.00
1007	54206	2023-04-21	3250.00
1008	54838	2023-04-30	1990.00
NULL	NULL	NULL	NULL

A join operation was performed between customer data and customer_returns table to identify churned customers who had also made complaints

- ❖ Display the return details along with the customer details of those who have churned and have made complaints.

```

6012 • SELECT
6013 cr.ReturnID,
6014 cr.CustomerID,
6015 cr.ReturnDate,
6016 cr.RefundAmount,
6017 cc.*
6018 FROM customer_returns cr
6019 JOIN customer_churn cc
6020 ON cr.CustomerID = cc.CustomerID
6021 WHERE cc.ChurnStatus = 'Churned'
6022 AND cc.ComplaintReceived = 'Yes';
6023

```

ReturnID	CustomerID	ReturnDate	RefundAmount	CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp
1002	50316	2023-01-23	2000.00	50316	0	Computer	2	29	UPI	Female	2
1004	52321	2023-03-08	2510.00	52321	18	Mobile Phone	3	19	E wallet	Male	2
1006	53749	2023-04-17	1740.00	53749	1	Mobile Phone	3	31	Credit Card	Female	3

NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplainReceived	ChurnStatus
3	Fashion	5	Married	10	15	1	1	2	200	Yes	Churned
1	Grocery	3	Single	7	17	1	2	8	237	Yes	Churned
5	Laptop & Accessory	4	Single	2	24	3	11	11	171	Yes	Churned

Key Business Benefits

- Identifies major churn indicators.
- Helps improve customer retention strategies.
- Enables targeted marketing campaigns.
- Improves customer satisfaction tracking.
- Assists management in reducing customer attrition.
- Supports data-driven decision making.

Conclusion

- This project successfully demonstrates how MySQL can be used to clean, transform, and analyse e-commerce customer data. The findings provide valuable insights into customer behaviour, churn patterns, complaint trends, and purchasing preferences, enabling businesses to develop effective retention strategies and improve customer loyalty.